

LAB ASSIGNMENT – 35

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35. Data Filtering and Focusing

Question: Given a sales dataset, filter records based on Sales greater than ■50,000, a particular region, and a particular product category.

Aim

To filter and focus on relevant records from a sales dataset using multiple conditions: sales greater than ■50,000, a selected region, and a selected product category.

Code

```
import pandas as pd

# Create a sample sales dataset
data = {
    "Order_ID": [101, 102, 103, 104, 105, 106, 107, 108],
    "Region": ["North", "South", "East", "South", "West", "South", "North", "South"],
    "Category": ["Electronics", "Furniture", "Electronics", "Electronics",
                 "Furniture", "Electronics", "Clothing", "Furniture"],
    "Sales": [45000, 62000, 75000, 85000, 40000, 95000, 55000, 70000]
}

df = pd.DataFrame(data)

# Apply all filtering conditions
filtered_data = df[
    (df["Sales"] > 50000) &
    (df["Region"] == "South") &
    (df["Category"] == "Electronics")
]

print("Filtered Sales Records:")
print(filtered_data.to_string(index=False))
```

Output

Order_ID	Region	Category	Sales
104	South	Electronics	85000
106	South	Electronics	95000

The output contains only those records where Sales is greater than ■50,000, Region is South, and Category is Electronics.

Result

Thus, the sales dataset was successfully filtered using multiple conditions, and the required records were obtained.